

Bio-Pulse Symbiosis

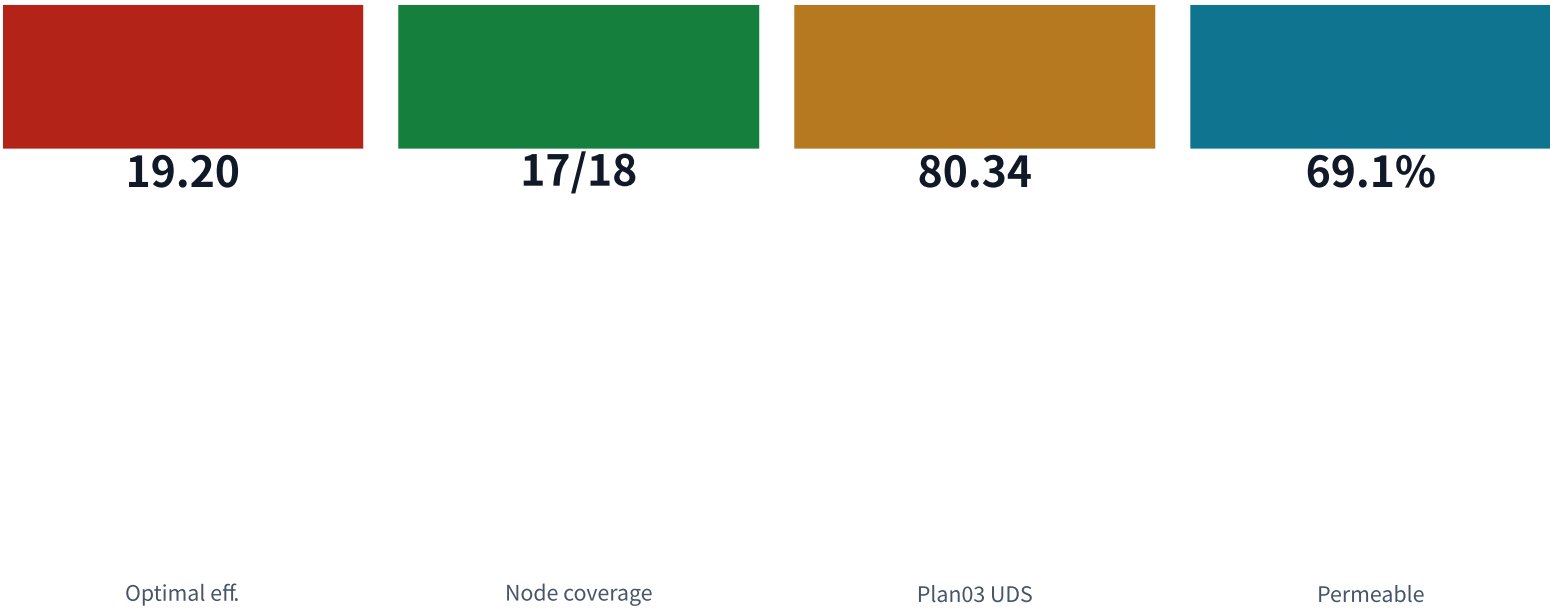
智脉共生



Bio-Physarum Mobility Network Renewal for the Dazhongsi AI Industry Cluster

Method: Bio-Physarum (Tero et al. 2010) + NSGA-II

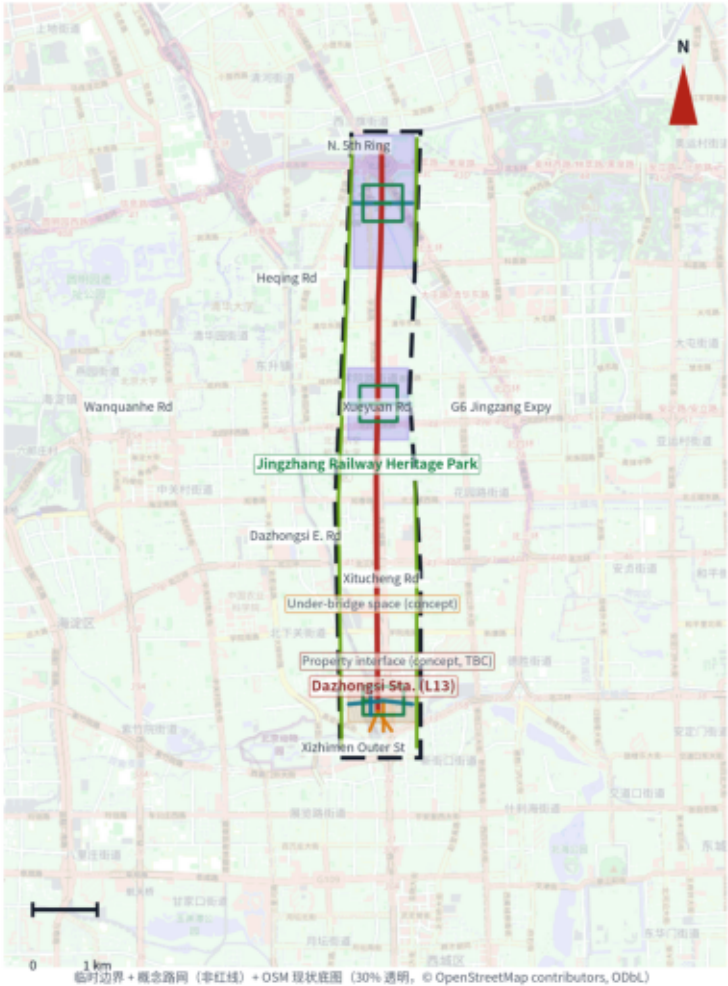
Validation: 167-edge / heritage f3≡f2



Recommended: Plan03 UDS 80.34 • permeable 69.1% • green penetration 25.2%

Formal geometry = conceptual network inside provisional boundary; real Physarum is method-validation evidence only.

Site Overview & Conceptual Network

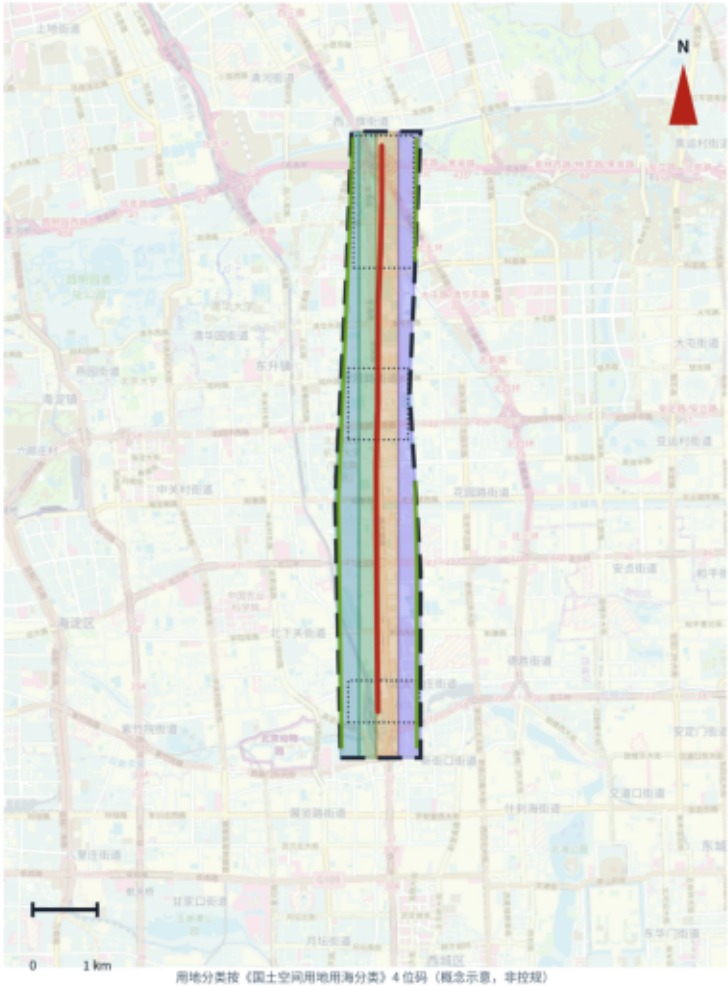


图例 Legend

- 粘菌主静脉 primary vein
- 次级支脉 secondary branch
- 慢行环 local loop
- 京张绿廊 green corridor
- 站点接驳 station feeder
- 众智园片区
- 北京AI原点社区
- 大钟寺AI产业集聚区

边界/路网/片区为算法模拟概念与临时范围（非实测、非审批几何）；道路/站点/公园名称基于 OSM 公开数据与公告文字四至（位置示意）；

Land-Use & Spatial Structure

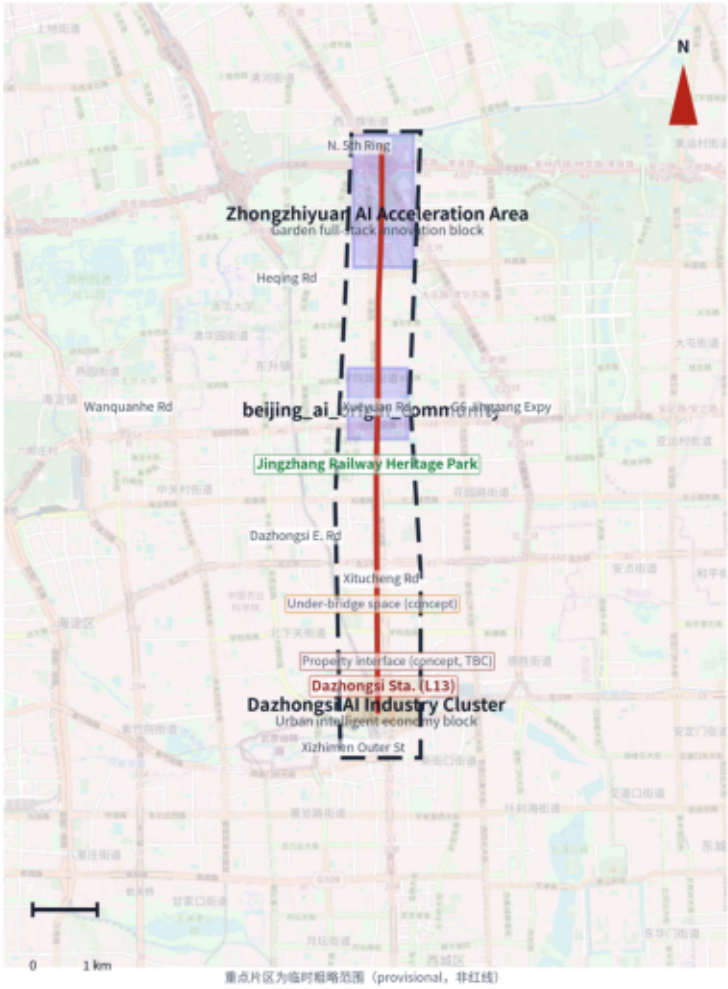


图例 Legend

- 0802 科研/研发用地 (AI研发试验)
- 1401 公园绿地与开放空间
- 05 商业商务与产业服务
- 0702 社区服务与居住
- 粘菌主静脉
- 京张遗址公园绿廊

用地色块与代码为《国土空间用地分类》概念示意 (agent_generated_design)，非控规用地、非红线；虚线为三处重点片区临时范围。

Three Key Areas + Physarum Spine

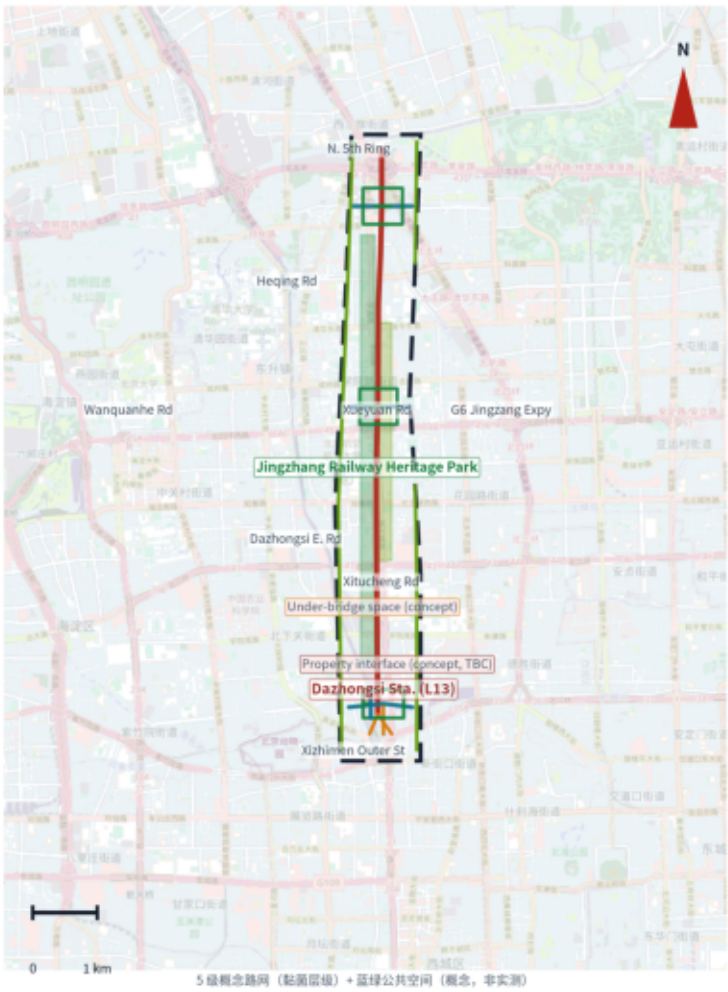


图例 Legend

- 众智园AI自主创新加速区
- 北京AI原点社区
- 大钟寺AI产业集聚区
- 黏菌主静脉 primary vein
- 场地临时边界（虚线）

三处片区边界为公告面积+文字四至推定的临时范围，非官方 polygon、非红线；站点坐标为 OSM 公开数据。

Mobility & Blue-Green Public Space

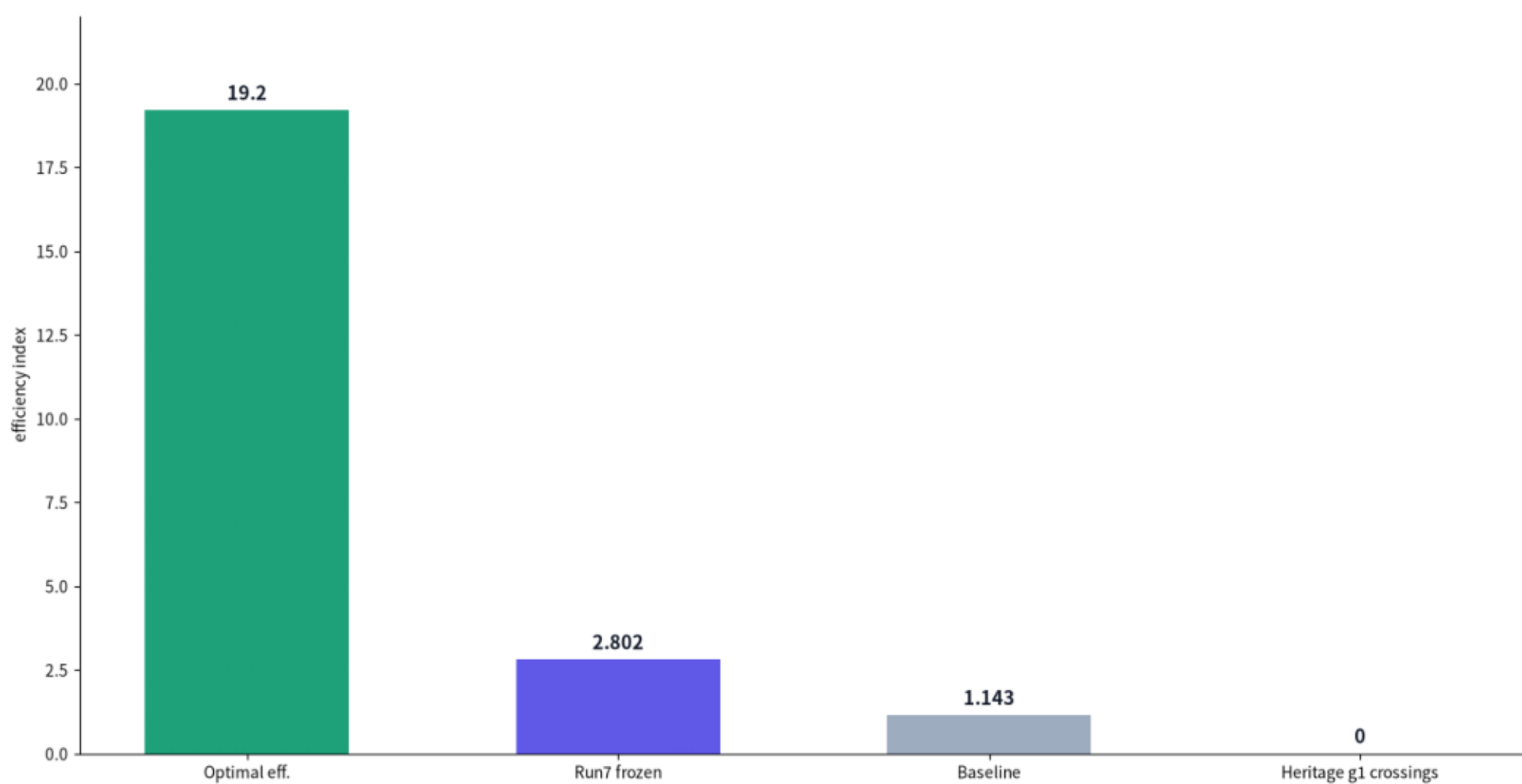


图例 Legend

- 主静脉 primary vein
- 次级支脉 secondary branch
- 慢行环 local loop
- 绿廊 green corridor
- 站点接驳 station feeder
- 公园绿地 green space
- 公共空间 public space

路网为 Bio-Physarum 模拟生成的概念线网（非审批红线，非实测几何）；蓝绿空间为 agent_generated_design 概念示意。

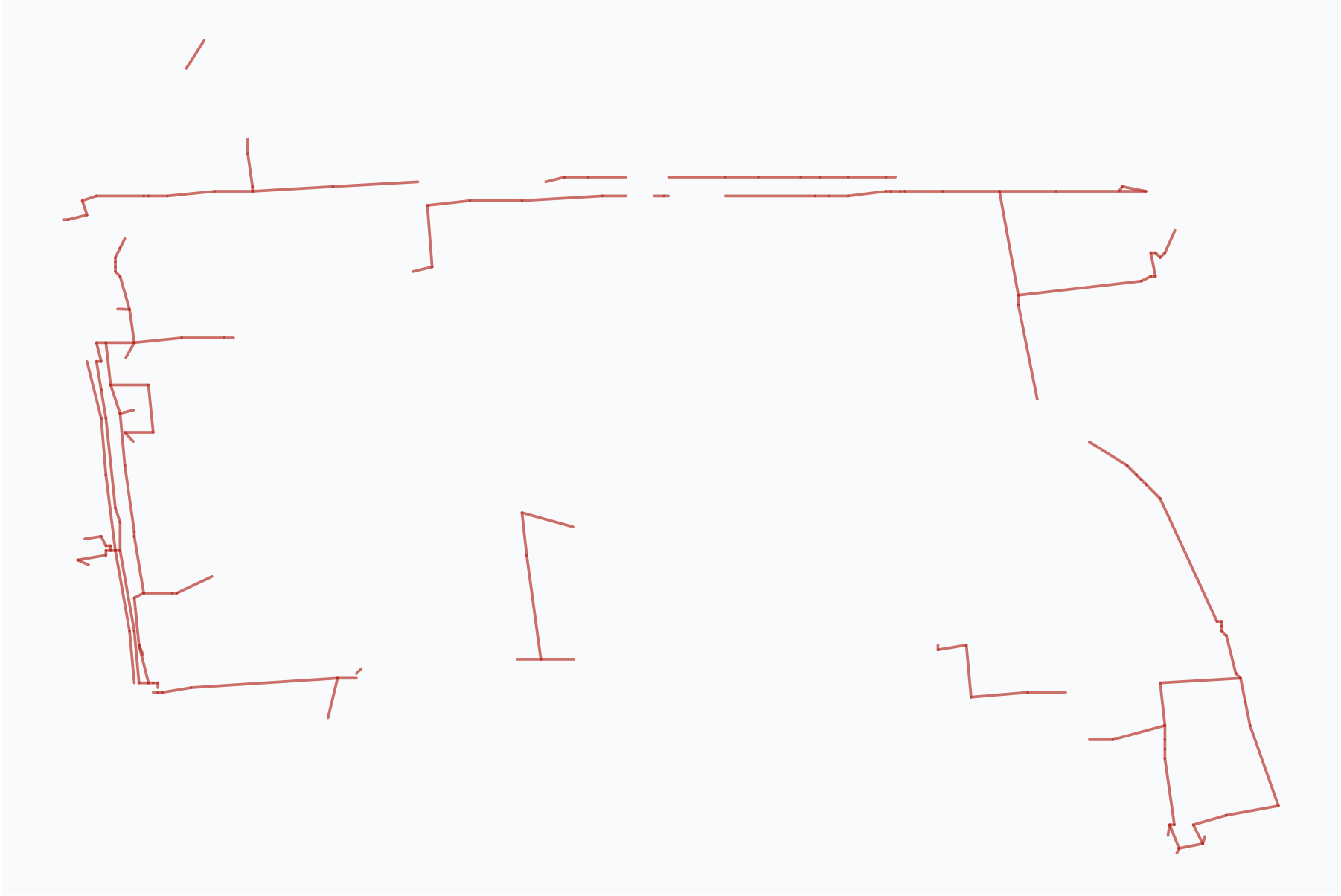
Metrics and method validation



Plan03 urban-integration UDS 80.34 recommended; heritage objective $f_3 \equiv f_2$ degenerates to cost — '0 heritage crossings' is g1 hard-constraint behavior only, not an independent site heritage conclusion
Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective $f_3 \equiv f_2$ (degenerates to cost; no independent site heritage conclusion)

Brand: 智脉共生 (Bio-Pulse Symbiosis)

Real Physarum 167-edge skeleton (method validation)



Zhi = AI belt Mai = Physarum network Symbiosis = blue-green • slow • industry • community

Conceptual brand (suggestive framework); does not replace existing names/trademarks/marks.

Visual base: real Physarum run skeleton (167 edges, ~2-3 km west of provisional boundary, method validation only).

Global Benchmark Cases (Public Information)

案例 Case	地点 Location	核心做法 (公开报道)	借鉴 / 警示
Sidewalk Labs Quayside	Toronto, CA	Sensing, adaptive building, data trust (2017-2020)	Governance mechanism / privacy caution
Masdar City	Abu Dhabi, UAE	Low-emission block, passive cooling, early PRT	Low-carbon + feeder / fiscal balance
Songdo IBD	Incheon, KR	Reclaimed town, U-City sensing, pneumatic waste	New infra reservation / keep organic growth
Amsterdam Smart City	Netherlands	PPP collaboration, living-lab pilots	Incremental + multi-party / responsibility
Xiong'an New Area	Hebei, CN	CIM twin, blue-green network, utility tunnels	Digital twin + blue-green slow priority

Test Validation Scenarios (suggestive framework)

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场景	验证对象	检验依据（真实）	关键目标
T-01 Dazhongsi TOD	Four-quadrant link & feeder	GB 50763-2012 / GB/T 51328-2018	Barrier-free flow; feeder on primary vein
T-02 Xiaoyue stormwater	Blue-green + permeable paving	Sponge City guide / GB 50014	Permeable 69.1% / green pen. 25.2%
T-03 AI Origin access	Campus-park-block stitching	GB 50763-2012	Slow loop covers nodes; continuous access

Implementation Matrix (suggestive framework)

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项目	建议责任主体	资金来源（建议性）	建议周期	暂停/退出条件
JZ-01 Slow-travel stitching	Transport commission + park	Fiscal + renewal fund	Near-term	Pause if ownership unsettled
JZ-02 Qinghe interface	Water authority + park	Fiscal + green bond	Mid-term	Exit if flood unmet
JZ-03 Conversion street	University + sci-tech bureau	University + social capital	Mid-term	Pause if ownership dispute
JZ-04 Dazhongsi four-quadrant	Rail operator + transport	Rail + fiscal	Mid-term	Postpone if timing mismatch
JZ-05 Edge computing	Sci-tech bureau + operator	Social capital + subsidy	Long-term	Exit if operator absent
JZ-06 Physarum recalibration	Design team + university	Research funding	Long-term	Keep 'to be confirmed'

Responsible bodies, funding, cycles are suggestive; not official approval or funding commitment.

Accessibility & Inclusivity (GB 50763-2012)

画像	无障碍与包容性需求	空间响应
Open-source developer	Night collaboration reachability & safety	Barrier-free entrance/toilet, graded lighting
Startup team	Members with disabilities	Barrier-free office/test space, lowered desk
Leading-enterprise visitor	International / reduced-mobility	Multilingual signage, barrier-free elevator
Nearby resident	Elderly, children, disabled daily passage	Curb ramps, tactile paving, rest seats
University teacher/student	Disabled campus-park connectivity	Barrier-free slow stitching, experience point